Practical No. 5

Adapter Based Views

1. Develop an Android application using ListView with ArrayAdapter and implement item click handling.
2. Develop an Android application using GridView with Custom Adapter and image selection handling.
3. Develop an Android application implementing RecyclerView as a modern alternative to ListView.

1. ListView

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
android:padding="20dp">
<TextView
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="□ Course List"
android:textSize="28sp"
android:textStyle="bold"
android:gravity="center"
android:padding="15dp" />
<ListView
android:id="@+id/listView"
android:layout_width="match_parent"
android:layout_height="match_parent" />
</LinearLayout>
```

MainActivity.java

```
package com.example.listviewdemo;

import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    ListView listView;

    String[] courses = {
            "B.Sc. Information Technology",
            "B.Sc. Computer Science",
            "BCA",
            "MCA",
            "M.Sc. Computer Science",
            "M.Sc. Information Technology"
    };

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        listView = findViewById(R.id.listView);

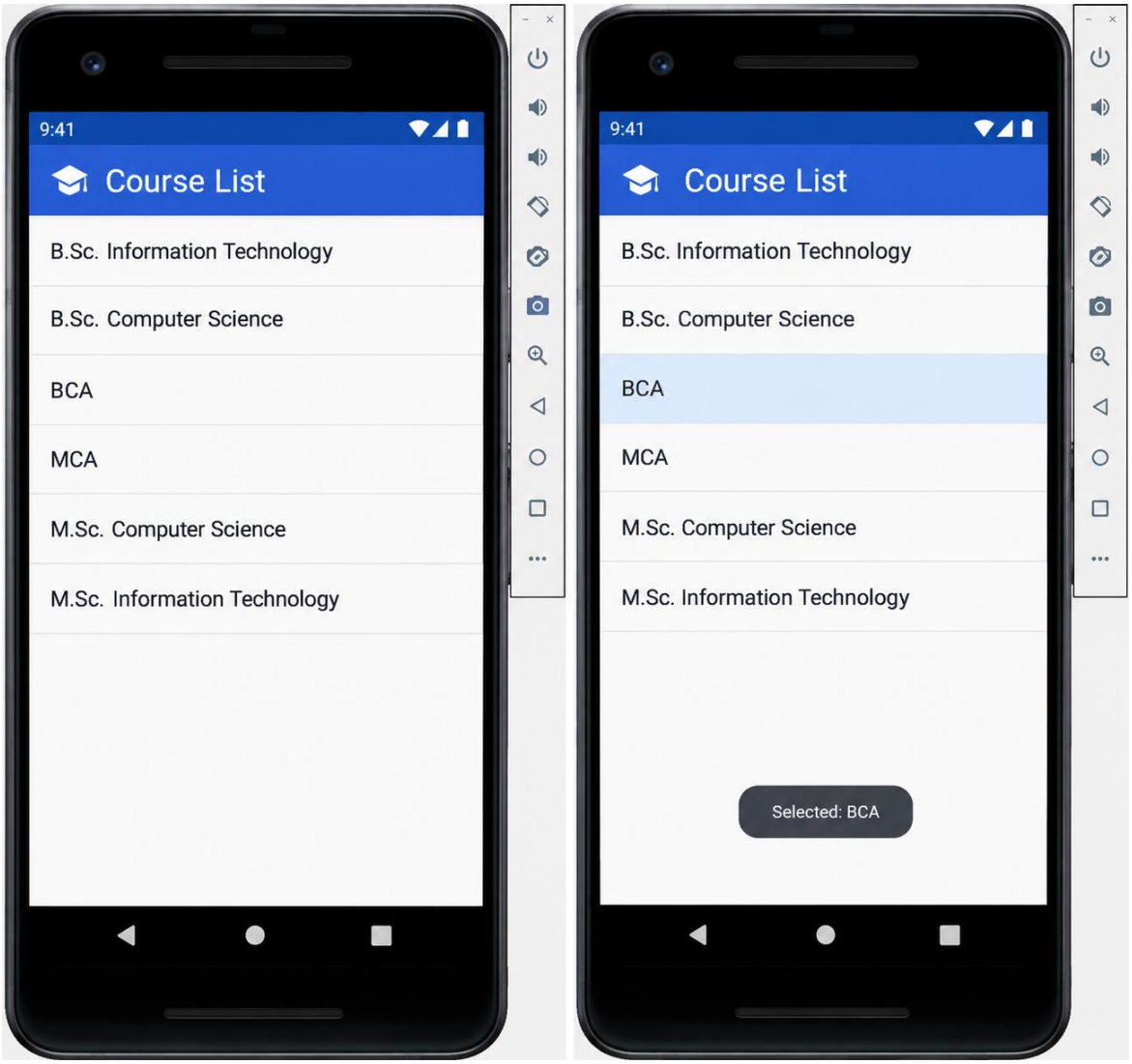
        ArrayAdapter<String> adapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_list_item_1,
                courses
        );

        listView.setAdapter(adapter);

        listView.setOnItemClickListener((parent, view, position, id) -> {

            String selectedCourse = courses[position];

            Toast.makeText(
                    MainActivity.this,
                    "Selected: " + selectedCourse,
                    Toast.LENGTH_SHORT
            ).show();
        });
    }
}
```


9:41
Course List
B.Sc. Information Technology
B.Sc. Computer Science
BCA
MCA
M.Sc. Computer Science
M.Sc. Information Technology
9:41
Course List
B.Sc. Information Technology
B.Sc. Computer Science
BCA
MCA
M.Sc. Computer Science
M.Sc. Information Technology
Selected: BCA